ORTHOGONALITY · PROJECTIONS

Properties of orthogonal projections

Two conditions define the whole object, and everything below them is a consequence. Idempotence alone gives a projection; symmetry is what makes it the perpendicular one.

07
PROPERTIES

UNCONDITIONAL 4

i Idempotence $P^2 = P$

UNCONDITIONAL

REQUIRES defining condition

Projecting twice changes nothing — once a vector is in W it stays put, so every vector of W is a fixed point of P. This is unusual among linear transformations, most of which keep changing a vector on repeated application.

ii Symmetry $P^\mathsf{T} = P$

UNCONDITIONAL

REQUIRES defining condition

Self-adjoint with respect to the dot product, meaning $P\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot P\mathbf{v}$ for every pair. This is the condition that makes the projection orthogonal rather than merely a projection — it forces the residual to be perpendicular to W, not just outside it.

v Complementary projection $I - P$ projects onto $W^\perp$

UNCONDITIONAL

REQUIRES P orthogonal

The residual is itself a projection, onto the orthogonal complement. Check: $(I-P)^2 = I - 2P + P^2 = I - P$ by idempotence, and $(I-P)^\mathsf{T} = I - P$ by symmetry — so it satisfies both defining conditions.

vii Eigenvalues $\lambda \in \{0, 1\}$

UNCONDITIONAL

REQUIRES follows from $P^2 = P$

From $P\mathbf{v} = \lambda\mathbf{v}$ and idempotence, $\lambda^2 = \lambda$, so λ is 0 or 1. The eigenvectors for 1 span W, those for 0 span $W^\perp$, and $\operatorname{rank}(P) = \operatorname{tr}(P) = \dim W$ — the trace counts the ones. See spectral properties.

CONDITIONAL 2

iv Orthogonal decomposition $\mathbf{b} = P\mathbf{b} + (\mathbf{b} - P\mathbf{b})$

CONDITIONAL

HOLDS WHEN P orthogonal — both conditions

Every vector splits uniquely into a part in W and a part perpendicular to it. Uniqueness is what makes this a decomposition rather than merely one way of writing $\mathbf{b}$, and it is the reason W and $W^\perp$ intersect only at the origin.

vi Minimal distance $\|\mathbf{b} - P\mathbf{b}\| \le \|\mathbf{b} - \mathbf{w}\|$ for all $\mathbf{w} \in W$

CONDITIONAL

HOLDS WHEN equality iff $\mathbf{w} = P\mathbf{b}$

The projection is the closest point of W to $\mathbf{b}$, and the only one achieving the minimum. This is the entire content of least squares — the best approximation is a projection, and the perpendicularity of the residual is why no other point can do better.

FAILS OR UNDEFINED 1

iii Oblique projection $P^2 = P$ but $P^\mathsf{T} \neq P$

NOT ORTHOGONAL

FAILS WHEN symmetry is dropped

COMMON ERROR treating idempotence alone as the definition of a projection onto W

Still a projection, and still onto the same subspace — but along a slanted direction rather than the perpendicular one. The two conditions are independent, and this entry exists to show that the second is not implied by the first.

WITNESS

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P = [[1, 1], [0, 0]]
P² = P, so it projects — onto the x-axis
Pᵀ ≠ P, so it projects along the line y = -x, not perpendicular
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Drop symmetry and P still projects onto W, but along some other direction — an oblique projection. The residual is then no longer perpendicular, the distance is no longer minimal, and least squares loses its justification entirely. That is how much rests on the second condition.